

Off the Rails: Examining Reliability and Ridership on the MBTA

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[GitHub Repository URL](#)

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Abstract

Public confidence in the Massachusetts Bay Transportation Authority (MBTA) has declined in recent years due to frequent service disruptions, slow zones, and reliability concerns. Riders and policymakers often debate whether these operational issues meaningfully shape ridership patterns, but the discussion is rarely grounded in data.

In this project, we set out to predict MBTA ridership using operational and temporal features, with a particular focus on disruptions such as slow zones and service alerts. We combined ridership data with detailed records of service performance from January 2023 through May 2024, producing more than 2,000 daily observations across the system's four rapid transit lines.

Our analysis shows that disruptions are consistently among the strongest predictors of ridership, especially when combined with weekend service patterns. These findings highlight that slow zones and alerts are not just anecdotal frustrations but measurable drivers of ridership decline. For system planners, the implication is clear: reducing disruptions is likely to yield real gains in restoring rider trust and usage.

Background and Question

Reliable public transit is essential for the daily life of Boston residents. The Massachusetts Bay Transportation Authority (MBTA), which operates the region's rapid transit system, has been under intense scrutiny for service disruptions ranging from slow zones to frequent alerts and delays. In 2023, for example, nearly one-third of

MBTA track mileage was under speed restrictions, drawing widespread media attention and frustration from riders who faced prolonged commutes (Boston Globe, 2023). Public trust in the MBTA has eroded, with ridership recovery lagging behind peer systems after the pandemic.

Academic and industry research consistently points to reliability as one of the most important drivers of transit ridership. Studies have shown that when service is unreliable, riders are more likely to switch modes or reduce their trips altogether. In Boston, this issue is particularly pressing given the availability of alternatives such as driving, cycling, and working from home.

Research Question

What factors best predict MBTA ridership, and how strongly do service disruptions like slow zones and alerts contribute compared to other predictors?

Hypothesis

We expect that disruptions will emerge as significant predictors of ridership, even when accounting for other temporal and operational factors.

Data

Our analysis relied on publicly available data from the Massachusetts Bay Transportation Authority (MBTA), which provided daily records of operational performance and passenger ridership across the Red, Orange, Blue, and Green Lines. These data sources allowed us to connect service disruptions, such as slow zones and service alerts, with outcomes in ridership and reliability. By working at the daily level across multiple rapid transit lines, we were able to capture both system-wide trends and line-specific variation.

The dataset covered the period from January 2023 through May 2024, resulting in just over 2,000 daily observations with 13 primary variables. Key measures included estimated ridership, on-time performance (OTP) scores, the number and severity of slow zones, and service alerts tagged by effect (delay, shuttle bus replacement). Additional contextual variables such as day of the week, month, and special events were also included to help capture expected variation in demand.

This dataset was well-suited to our research question. It contained both the operational indicators most likely to impact riders (slow zones and alerts) and direct measures of passenger response (ridership). While some limitations were present, including missing dates in certain alert files and potential inconsistencies in how alerts were logged, the breadth and daily granularity of the MBTA's reporting made it an appropriate foundation for modeling the relationship between disruptions and transit outcomes.

Data Acquisition

We drew on several publicly available data sources maintained by the Massachusetts Bay Transportation Authority (MBTA). Daily ridership data came from MBTA's automated passenger count system, which reports estimated station entries by rapid transit line. Reliability measures were obtained from the MBTA's on-time performance (OTP) dashboard, which provides a daily expected passenger wait times vs actual passenger wait times, which can be divided to obtain an OTP score between 0 and 1. Information on track slow zones was collected from the MBTA's published slow zone tracker, which records the number of active restrictions, the percentage of track affected per rapid transit line, and total miles under restriction. Service alert data were sourced from MBTA's real-time alerts feed and included the type of disruption (such as delays or shuttle replacements) and its timing.

Together, these sources gave us a dataset that captured both the causes of service disruption (slow zones, alerts) and the consequences (ridership, reliability). We aggregated the data into a single dataset with daily records for each of the four rapid transit lines: Red, Orange, Blue, and Green, covering January 2023 through May 2024. The final merged dataset contained 2,056 daily observations with 13 variables before additional feature engineering.

This combination of sources was appropriate for our research question because it tied operational factors most likely to affect passenger experience directly to outcomes in system performance and usage. While MBTA data are not perfect for example, some service alert records were missing dates or contained duplicates. They represent the most detailed and comprehensive view currently available to the public of how disruptions unfold on the rapid transit system.

Data Cleaning

Once the ridership, reliability, slow zone, and service alert datasets were merged, several cleaning steps were required before modeling. First, we standardized the service date column to datetime format and sorted the records by transit line and date. This was essential for generating lagged features and respecting the time-series nature of the data. Some data came in half-hour increments (gated station entries) and needed to be aggregated to daily counts in order to be merged with other data.

One challenge came from the service alert files. Some rows were missing dates, which caused near-duplicate records to overwrite valid ones when duplicates were dropped. We corrected this by removing rows without dates prior to duplicate removal. This adjustment improved data integrity, reduced skewness in alert-related variables, and ultimately increased model performance.

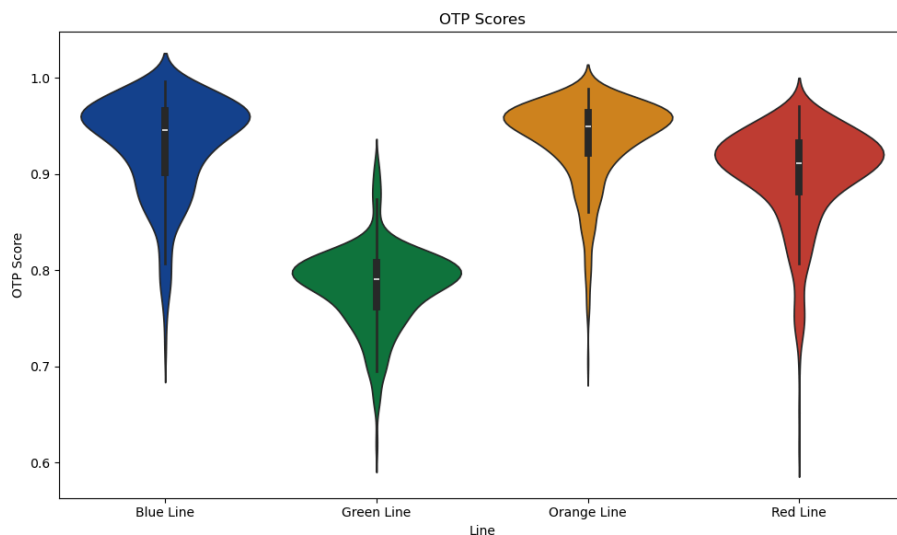
Next we addressed skewness in numerical variables. Highly skewed variables such as number of alerts and delay-related alert effects were log-transformed, while moderately skewed features such as number of slow zones were square-root transformed. These transformations brought distributions closer to normality, making them more suitable for regression-based models.

Categorical variables, including line, day of the week, and month, were one-hot encoded. This expanded the dataset from 13 to 38 columns but ensured that categories were treated appropriately without imposing artificial order. After cleaning, the dataset contained 2,056 daily observations from January 2023 through May 2024, with all

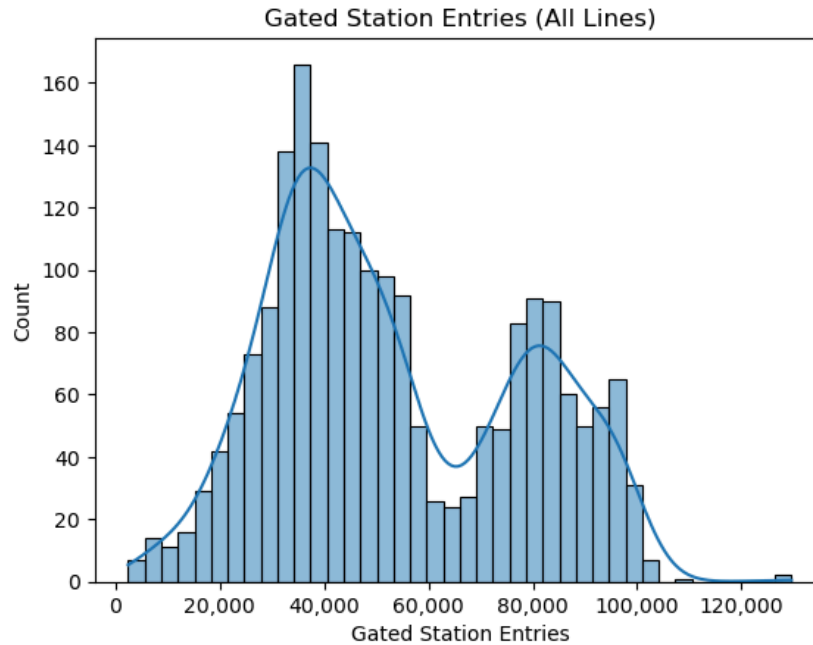
variables formatted correctly, no missing values, and a structure ready for feature engineering and model training.

Data Exploration (EDA)

Exploratory analysis was conducted on the cleaned dataset to understand distributions, relationships, and potential issues prior to modeling. Ridership showed a strong right skew, with daily entries ranging from fewer than 15,000 on some lines to over 60,000 on others. This justified the later use of transformations and lag features to stabilize variation. Reliability (OTP scores) clustered near 1.0 on most days but varied more widely on lines with heavier slow zone activity, such as the Red Line.



Violin plot showing distributions of on-time performance scores for each rapid transit line.



Histogram showing distribution of gated station entries across all rapid transit lines, with a right skew.

Slow zones and alerts exhibited skewed distributions with frequent low values but occasional extreme spikes. These spikes aligned with periods of major disruptions, such as long-term closures, which suggested they could heavily influence model predictions.

Correlation analysis indicated that ridership was negatively associated with both the number of slow zones and the share of track under restriction. OTP scores were positively correlated with ridership, reinforcing the idea that reliability matters for demand. However, multicollinearity was evident among the three slow zone variables (number, percent of track, miles affected) and among the different alert measures. This motivated the use of Principal Component Analysis (PCA) in preprocessing.

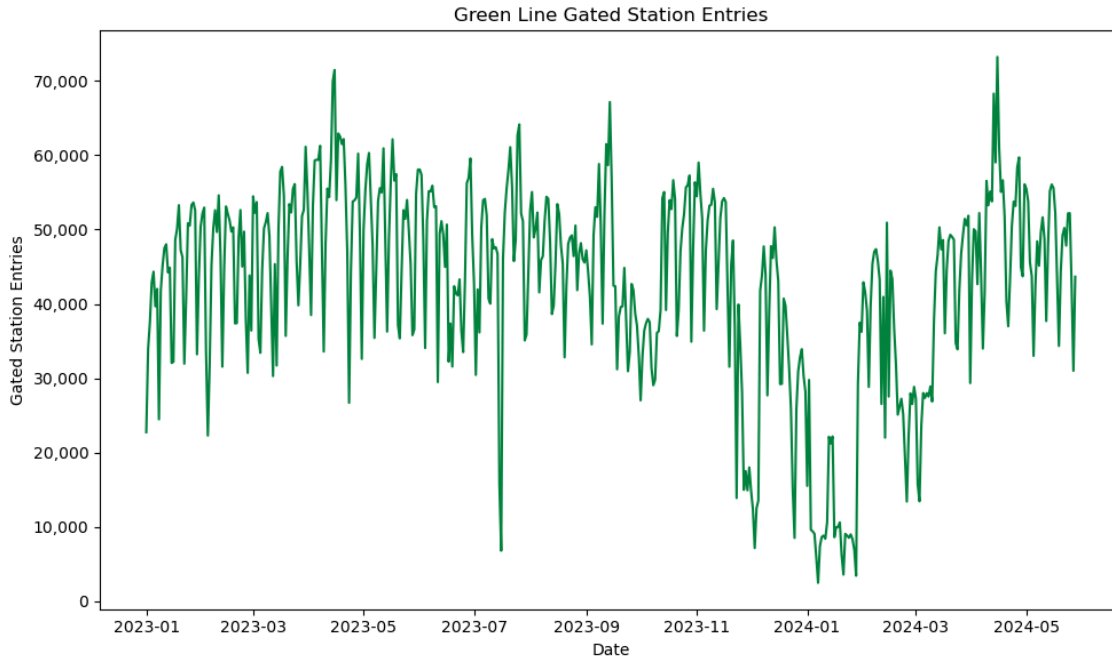
	num_slow_zones	total_track_pct	total_miles_affected
num_slow_zones	1.000	0.100	0.924
total_track_pct	0.100	1.000	0.352
total_miles_affected	0.924	0.352	1.000

Slow zone variables were highly correlated.

	num_alerts	alert_effect_delay	alert_effect_shuttle
num_alerts	1.000	0.812	0.502
alert_effect_delay	0.812	1.000	0.234
alert_effect_shuttle	0.502	0.234	1.000

Alert-related variables were highly correlated.

Visual inspection of time series plots confirmed distinct ridership dips during major closures and weekends, underscoring the need to include both temporal indicators (day of week, month) and service disruptions as predictors. These EDA findings directly informed our feature engineering, particularly the creation of lag variables, rolling averages, and composite disruption measures.



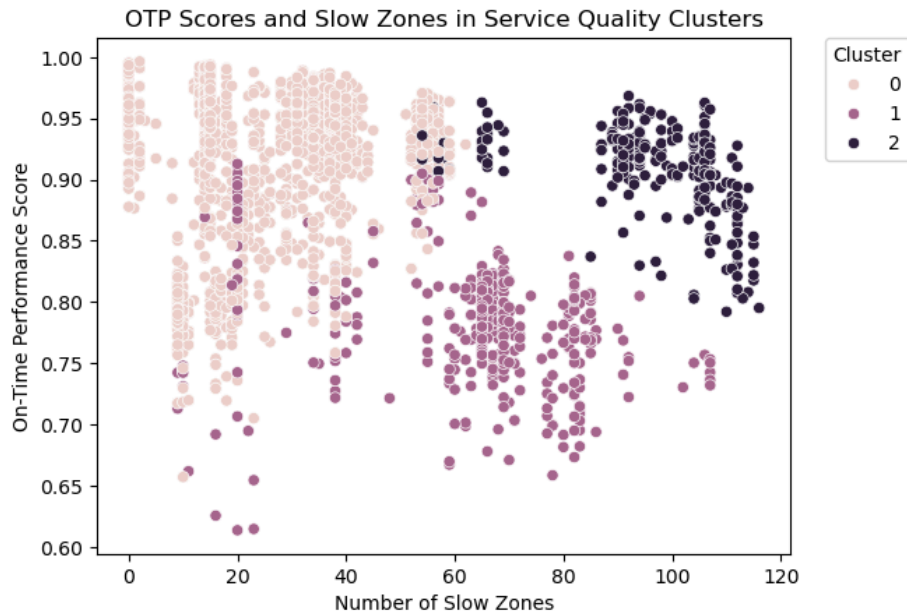
The Green Line shows an example of a distinct dip in gated station entries during widespread closures in January of 2024 (DeCosta-Klipa, 2024).

Pre-processing and Feature Engineering

We observed multicollinearity between a couple sets of variables. The first set was the number of slow zones, total miles affected by slow zones, and the total percentage of track affected by slow zones. For this, we employed Principal Component Analysis, which would address both multicollinearity and dimensionality reduction. We kept two infrastructure stress principal components which together explained 98.6% of the variance in the variable set. The second set of variables with significant multicollinearity was number of alerts, number of alerts resulting in a delay, and number of alerts resulting in shuttle replacements. We again kept two inconvenience principal components which together explained 95.4% of the variance. Both times, the PCA was

fit on only our training set (the entire year of 2023) and applied to both the training and test sets to avoid data leakage.

We also saw an opportunity to use K-means clustering to assign a service quality cluster to each observation. We used the on-time performance score, number of slow zones, number of active alerts, and one of our infrastructure stress principal components to cluster the data. This resulted in the creation of three service quality clusters. The first cluster (cluster 0) represents days with a higher average on-time performance score (0.916), a lower average number of slow zones (24.2), and a lower average number of active alerts (1.3). This represents a good operational day. The second cluster (cluster 1) represents the lowest average on-time performance score (0.773), a medium average number of slow zones (68.9), and the higher average active alerts (3.6). The final cluster (cluster 2) represents a higher on-time performance score (0.906), but higher average slow zones (98.1) and alerts (3.4). Cluster 0 makes up 886 observations in the training set and 528 in the test set. Cluster 1 makes up 336 observations in the training set and 57 in the test set. Finally, cluster 2 makes up 238 observations in the training set and 11 in the test set. The proportion of good operational days in our test set is much higher than in the training set.



We also created new variables which we believed could contribute to the analysis. First, we created lagged variables in order to capture temporal effects on outcome variables, where one day's performance may have an effect on the next. The lagged variables include a 1-day lag for on-time performance score and ridership, as well as 7-day averages for number of slow zones and on-time performance.

We developed interaction terms to capture how operational problems affect different time periods and lines differently. Key interactions included slow zones crossed with weekend indicators and alerts crossed with weekend flags, recognizing that service disruptions may have varying impacts during different usage patterns. We also created line-specific interactions with slow zones to capture how each transit line responds differently to infrastructure problems.

Algorithms Selection

We implemented three supervised models to predict MBTA ridership: Ordinary Least Squares (OLS), Lasso regression, and Random Forest regression. Each was chosen for different strengths, and together they gave us a balanced view of the predictive performance, interpretability, and model flexibility.

Ordinary Least Squares

We first fit an OLS model on the data. This involved first making sure we did not include any variables we already performed PCA on as well as dropping one of each one-hot encoded variable categories. We also dropped one of each pair of highly-correlated variables ($R^2 > 0.9$). The OLS model was used as a baseline model to compare to other models. Upon viewing residuals, which appeared to take a U-shape, and upon conducting a Jarque-Bera test, we concluded that the residuals were not normally distributed. With this model we achieved a test R^2 of 0.61 and an MSE of 246,083,030.728.

Lasso Regression

The next model we fit was a Lasso regression model. Again, we made sure not to use any features we had already used to form principal components and dropped one of each one-hot encoded variable categories (indicators for Blue Line, Sunday, January, and service quality cluster 0). Lasso stands up better against multicollinearity, so we did not drop one of each pair of highly correlated variables. We standardized the predictors in both the test and training sets by fitting on the training set and then applying the standardization to both. This would prevent any variables with larger coefficients from dominating the analysis.

For this model in order to control for overfitting, we used scikit-learn's LassoCV() function and TimeSeriesSplit() cross-validation folds as well as tuned for the best alpha. The TimeSeriesSplit() cross-validation folds compare one time period against the next, then the combination of those time periods against the next after that, and so on. We found that the best alpha was 37.326. Scikit-learn will give larger alphas like this, and it can be made comparable to alphas used by other libraries by dividing by the number of observations. This yields a comparable alpha of 0.026. When predicting against the training set our R^2 was ~ 0.84 , and our test R^2 was ~ 0.81 , so our efforts to control overfitting kept these values somewhat close.

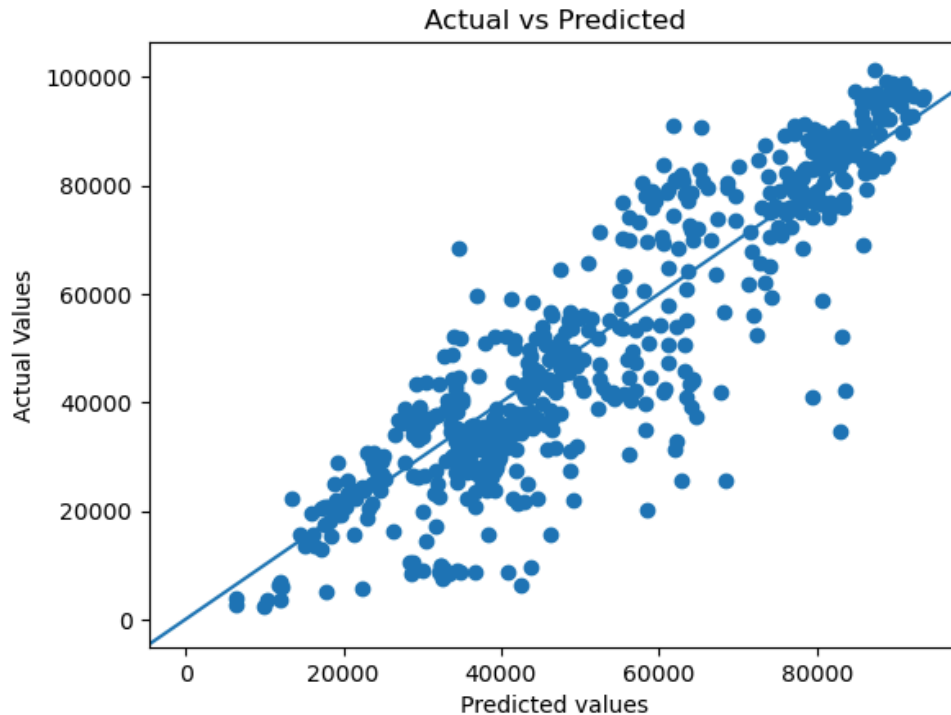
Random Forest

Finally, we used Random Forest regression to capture non-linear effects and complex interactions. This model was especially useful because it does not require strong statistical assumptions like normality of residuals. We tuned hyperparameters using grid search with TimeSeriesSplit cross-validation. The parameters tested included number of trees (100–500), maximum depth (4–20), and number of features per split (sqrt or log2). Scaling was not required due to the tree-based structure. Random Forest delivered strong overall performance, with a test R^2 of 0.80 and comparable MSE across models. Cross-validation results were consistent, suggesting the model was not overfitting. Feature importance rankings highlighted service alerts, slow zones, and lagged ridership as the strongest predictors, aligning with our expectations about disruptions driving ridership patterns.

Final Model

Results

Our strongest model was the Lasso regression model, which achieved an R^2 of ~ 0.81 .



Predicted vs actual values, from the Lasso regression model. The model achieved $R^2 = 0.81$.

Interpretation

Since we dropped one of each of 4 one-hot encoded variables, these serve as baselines: January serves as a baseline for other months, the Blue Line serves as a baseline for other train lines, service quality cluster 0 serves as a baseline for other service quality clusters, and Sunday serves as a baseline for other days of the week. Unsurprisingly, the Red Line variable had a strong positive coefficient because it had stronger ridership than the Blue Line.

The strongest predictor of ridership was the one-day ridership lag variable, with a coefficient of ~11,393, meaning that one standard deviation change in the lagged variable is associated with an 11,393 increase in ridership. We also saw the emergence of some of our interaction variables among significant predictors. The weekend flag and the weekend and slow zone interaction variables has coefficients of -3,169 and -6,693 respectively, indicating that while weekends are associated with drops in ridership, weekends with greater numbers of slow zones are associated with even more significant drops. Service quality clusters 1 and 2 had coefficients of -579 and -913, meaning that compared to service quality cluster 0 (good operational days), they appear to be associated with drops in ridership. These clusters represented day/line combinations with middle-to-higher numbers of slow zones.

One surprising outcome was the reliability and slow zone interaction term having a higher coefficient of 5,909. One possible explanation for this is that the Red Line tended to have higher reliability scores and the highest median ridership despite having more slow zones than other lines in 2023. Another surprising finding was that the interaction term between orange line and slow zones had a positive coefficient. This could be because the orange line is popular.

Otherwise, we see that some months have higher or lower typical ridership than January, people tend to use the train more on weekdays, and OTP scores make an appearance among predictors with positive coefficients, meaning higher OTP scores may be associated with higher ridership. See the appendix for all coefficients assigned to variables, including variables with coefficients reduced to 0.

Conclusion

This project set out to identify the strongest predictors of MBTA ridership and to determine how service disruptions such as slow zones and alerts contribute compared to other operational and temporal factors. Using over 2,000 daily observations from January 2023 through May 2024, we combined ridership and on-time performance data with records of track restrictions and service alerts across all four rapid transit lines.

Our modeling approach included baseline Ordinary Least Squares regression, regularized Lasso regression, and Random Forest regression. Together these models allowed us to balance interpretability with predictive accuracy while controlling for issues such as multicollinearity and overfitting. Exploratory analysis, dimensionality reduction through PCA, clustering, and the creation of lagged and interaction features provided a strong foundation for the models.

The results consistently showed that disruptions matter. Both Lasso and Random Forest identified service alerts, slow zones, and temporal interactions (such as weekends combined with disruptions) as among the most influential predictors of ridership. Lasso regression provided a compact, interpretable model with strong performance ($R^2 \approx 0.81$), while Random Forest captured complex non-linear patterns and delivered comparable predictive accuracy ($R^2 \approx 0.80$). These findings support our hypothesis that disruptions are not only visible in the data but also among the most important factors shaping daily ridership.

By linking operations directly to ridership outcomes, this analysis provides evidence that reducing disruptions is likely to have meaningful impacts on restoring both reliability and

rider confidence. The work also highlights the value of pairing traditional regression approaches with machine learning models to capture the full complexity of urban transit systems.

Discussion & Next Steps

The results of our models confirm that service disruptions such as slow zones and alerts are among the strongest predictors of MBTA ridership. This finding aligns with both rider experience and broader transportation research, underscoring that disruptions do not simply cause frustration but also translate into measurable decreases in daily station entries. The interaction effects we observed, particularly around weekends combined with higher disruption levels, suggest that the impact of operational problems is not uniform. Riders appear especially sensitive to disruptions when the service is already less frequent, which highlights how operational reliability interacts with normal demand cycles.

While our Random Forest and Lasso models performed well, there are still areas where our approach could be expanded. The Random Forest identified patterns that were difficult to interpret directly, while the Lasso emphasized clear predictors but could not fully capture the nonlinear complexity of the data. In practice, these models complement one another, but additional validation would further strengthen our conclusions.

Next steps should include testing additional models that are designed specifically for time series data. Autoregressive models with exogenous variables (such as ARIMAX) or

gradient boosting methods like XGBoost could provide both predictive power and stronger insights into temporal dynamics. Further hyperparameter tuning, particularly for tree-based models, could also help refine performance. Another important extension would be explicitly controlling for line-level fixed effects, since our results suggest that the Red Line, with higher slow zone activity, may be driving much of the observed relationship between disruptions and ridership. As more data becomes available, the analysis could also examine weekly or monthly trends instead of daily counts.

Line-level effects are one of the caveats that comes with this analysis. The Red Line example highlights this. Other caveats include the ridership data coming from gated station entries, with some train lines having non-gated stations. Also, low ridership during shutdowns does not necessarily represent people making the decision not to take the train, but it does represent the MBTA not collecting fare where they would have as well as inconvenience and disruption to passenger commutes.

Finally, while our project focused on ridership, applying the same methods to reliability as the outcome variable could provide a fuller picture. This would allow us to compare how operational disruptions translate first into reliability drops and then into ridership changes. By framing the analysis in this way, future work could provide stronger evidence of causal pathways, which would be valuable for both system planners and policymakers.

Code Availability

All code and processed datasets used in this project are available on our [GitHub Repository](#). The repository includes scripts for data cleaning, preprocessing, feature engineering, exploratory analysis, model fitting, hyperparameter tuning, and figure/table generation. This ensures full reproducibility of our results and transparency in the analysis workflow.

References

(2024, January 1). *MBTA is eliminating frustrating slow zones at record pace*. The Boston Globe. <https://www.bostonglobe.com/2023/12/29/metro/slow-zones-repairs>

DeCosta-Klipa, N. (2024, January 2). *Much of the Green Line will be closed (again) for most of January. here's what to know*. WBUR News. <https://www.wbur.org/news/2024/01/02/mbta-green-line-boston-january-shutdowns-slow-zones-newsletter>

Appendix

ridership_lag1	11393.040494
line_red	6129.345944
reliability_x_slow_zones	5924.437055
slow_zones_x_orange	4685.026717
day_of_week_Monday	2233.607135
day_of_week_Tuesday	1522.544283
line_green	1512.498669
day_of_week_Wednesday	708.005292
otp_score_7day_avg	417.641635
month_March	165.650028
day_of_week_Saturday	146.033181
month_October	125.398529
month_August	106.692221
month_September	88.488707
otp_score_lag1	85.869443
has_special_event	67.722009
inconvenience_pc1	65.481203
alerts_x_weekend	42.792718
month_July	25.315494

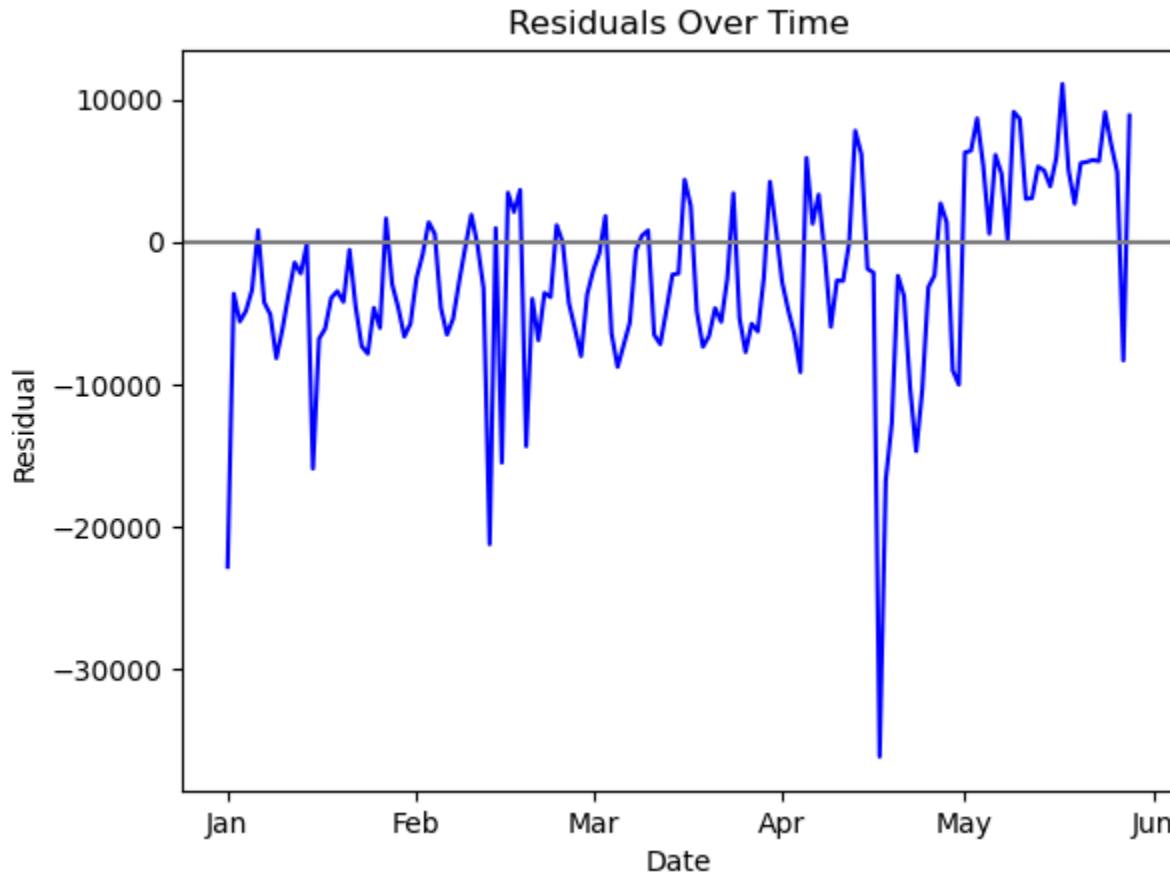
Variables with positive coefficients as a result of Lasso Regression. This shows months with higher ridership than January, days with higher ridership than Sundays, and other factors that may be associated with higher ridership.

month_February	-55.327761
month_May	-254.616009
month_April	-269.346189
service_date	-450.128768
month_November	-461.441307
inconvenience_pc2	-538.026096
service_quality_cluster_1	-579.478467
service_quality_cluster_2	-912.948594
month_December	-1113.976093
day_of_week_Friday	-1314.096410
infrastructure_stress_pc1	-1323.675540
weekend_flag	-3169.187920
slow_zones_x_weekend	-6692.556350

Variables with negative coefficients as a result of Lasso Regression, showing months associated with lower ridership than January, and other factors associated with lower ridership levels.

otp_score	-0.000000
slow_zones_7day_avg	-0.000000
slow_zones_x_blue	-0.000000
month_June	-0.000000
slow_zones_x_green	-0.000000
infrastructure_stress_pc2	0.000000
day_of_week_Thursday	-0.000000
line_orange	0.000000
total_track_pct_log	0.000000
slow_zones_x_red	0.000000

Variables whose coefficients were reduced to 0 as a result of Lasso Regression; they did not have significant predictive power.



Residuals over time, as a result of the Lasso Regression model. This shows the Blue Line as an example. There was a shutdown of many Blue Line stations that the model struggled to predict, depicted by a large negative residual that indicates that the model overpredicted ridership.